

than with legacy infrastructure wherein the storage and the servers are separated.

Eve: Due to consolidated layers, is the flexibility in choosing the suitable layers also reduced?

Bob: Not only is there reduced flexibility due to the use of pre-configured systems, there are other limitations also like vendor lock-in, the higher price for the infrastructure, etc., which makes this option suitable for organisations with certain fixed requirements from the application.

Eve: I have been hearing of one more option called the hyper-converged storage solution. Does it overcome the challenges faced in converged storage?

Bob: In hyper-converged storage, the underlying data architecture has been completely reinvented, allowing data management to be simplified. It also carries forward the benefits of convergence, including a single shared resource pool. Unlike converged systems where each component can be used for different purposes, in a hyper-converged storage solution, each component is used in a dedicated fashion for a specific purpose. Hyper-convergence goes far beyond servers and storage, bringing into the convergence fold many services and features including data protection features (backup and replication), de-duplication, wide area network (WAN) optimisation, solid state drive (SSD) arrays, SSD cache arrays, replication appliances or software, etc.

Stratoscale lets you build your own hyper-converged open stack cloud. Hence you can converge your networking storage and computing at a very low cost. Refer to this link <http://www.stratoscale.com/solutions/hyper-convergence/> for more information.

Eve: So these are the various on-premise storage options available. You had mentioned alternative cloud based solutions. How are they different from on-premise solutions in terms of features offered?

Bob: Cloud based storage solutions basically provide you various storage management services in a 'pay for what you use' fashion. They manage your data on a third party site and hence save you staffing and hardware costs as all the tasks are handled by service providers. Not only do they store and manage your data, a number of cloud storage services even provide you features like data redundancy, availability, scalability, etc.

Security is one of the biggest concerns but now there are cloud implementations (like private clouds) that will overcome any such risks. There are other downsides like service outages, limited control and flexibility, the risk of Internet based security breaches, vendor lock-in, etc., which might impact your applications and services.

Eve: I see... so which are the open source cloud storage solutions available?

Bob: I will just list out a few important and popular ones. OpenStack is a highly distributed, cost-effective option for building a cloud based storage solution and it has the huge

community support typical of an open source project. It offers a wide variety of storage services like scalability, redundant object storage, data integrity, replication, durability, the use of clusters to store petabyte data and many more features, about which you can learn more at <https://www.openstack.org/software/openstack-storage/>. According to OpenStack, the top 10 auto makers in the world use OpenStack to turn customer insights into action (<http://www.openstack.org/enterprise/big-data/>).

CloudStack was an alternative to OpenStack, launched by Citrix. But the monolithic architecture of CloudStack unlike the modular one of OpenStack limits scalability. Moreover, CloudStack supports only block based storage and hence is limited to specific application scenarios. For more information on CloudStack, visit <https://cloudstack.apache.org/>.

Ceph is another widely adopted open source storage solution. It is a unified storage solution that provides block, file as well as object storage capabilities. Other highlights of this tool are its scale-out feature, applicability to a variety of use cases due to the use of different hardware for different workloads, etc., rendering it ideal for enterprises as well. Visit <http://ceph.com/ceph-storage/> for more information on Ceph.

Eve: So, which of these would be the best data storage solution?

Bob: The choice very much depends on the exact application and customer requirements. For example, for unstructured data, object storage turns out to be the best solution. If you are providing services to customers, you can opt for building an on-premise cloud or provide a storage solution stack (converged or hyper-converged infrastructure) to your customer. In the cloud, if your customers have doubts or concerns about security, you can host a private cloud or hybrid cloud for a specific set of customers.

While all the storage solutions are aimed at reducing costs and making the set-up and deployment procedures as customer friendly as possible, the choice of the tool should only be made after analysing the organisation's business goals and the value addition provided by the tool's features, to potential customers.  **END**

Reference

http://www.snia.org/tech_activities/tech_center

By: Shubhra Rana, Anupama Sikchi and Vinay Patkar

Shubhra Rana is a software development engineer at Dell R&D, Bengaluru, and is interested in network security and cryptography.

Anupama Sikchi is a software development engineer at Dell R&D Bengaluru and is interested in networking and programming in Java.

Vinay Patkar works as a software development engineer at Dell India, R&D Centre Bengaluru and has an interest in cloud computing, data analytics and machine learning.